COFFEE PACKAGING QUALITY INSPECTION REPORT

Real-Time AI Seal Inspection and Physical Leak Detection

1. Report Information

Report ID	RPT-20260821-B9F982
Inspection Date	21 August 2026
Inspection Time	10:31:03 PM
Inspection Module	Packet Seal and Leak Quality Inspection

2. Overall Inspection Summary

Inspection	Result
Real-Time AI Overheat Inspection	OVERHEAT_DETECTED
Physical Packet Leak Test	GOOD
Overall Quality Decision	REJECT

3. Final Quality Decision

REJECT

Potential overheat defect was detected by the real-time AI inspection system.

4. Real-Time Two-Stage AI Inspection

Detected Seal Regions	2
Overheat Detected	YES
Highest Overheat Confidence	48.2%
AI Inspection Status	OVERHEAT_DETECTED

Seal-by-Seal Results

Seal	Seal Confidence	Overheat	Overheat Confidence
Seal 1	84.6%	NO	-
Seal 2	81.5%	NO	-

Real-Time Inspection Evidence


PACKET: OVERHEAT_DETECTED
SEAL 84.6% | OVERHEAT
SEAL 81.5% | OVERHEAT
over heat 48.2%
over heat 47.4%
over heat 27.0%
over heat 30.2%
Lakmee
Coffee
6g
Net Weight

5. Physical Packet Leak Detection

Leak Test Result	GOOD
Initial Value	300972
Average Value	282208.62
Threshold	286000
Minimum	281281
Maximum	282801
Range	1520
Reading Count	11
Final Device Status	TOP_REACHED

Load Cell Readings

Reading Number	Sensor Value
1	281281
2	281553
3	281706
4	281956
5	282136
6	282334
7	282431
8	282622
9	282706
10	282769
11	282801

6. Recommended Action

- Hold or reject the affected packet.

- Inspect sealing temperature settings.

- Verify sealing dwell time and heater condition.

- Inspect subsequent packets from the same batch.

7. Inspection System Information

AI Stage 1	Seal Region Object Detection

AI Stage 2	Overheat Defect Object Detection
Camera	IP Webcam Real-Time Video
Physical Controller	Arduino Uno
Force Measurement	HX711 Load Cell System
Mechanical Test	Motorized Packet Pressure Mechanism
Safety / Position Control	Top and Bottom Limit Switches

This report was generated automatically using the coffee packet quality inspection system.